Wireless Power Transmission: Designing a Near-Field System and Current Limitations of Far-Field Systems

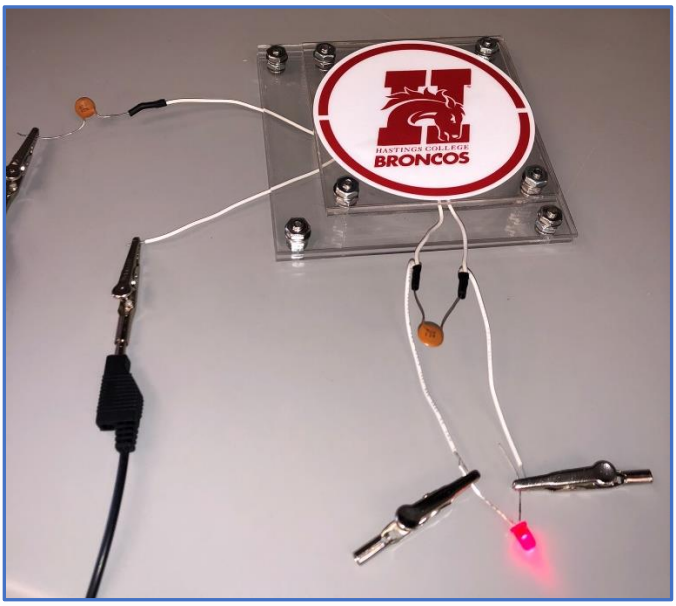


***Figure 1**: Resonant Inductive Coupling Design*

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Abstract

Innovations within the last decade have allowed for the average consumer to experience greater applications of wireless power transmission (WPT) in their daily life. Smartphones, watches, and even some electric vehicles feature WPT called near-field transmission (NFT) that is widely available to consumers today. Nikola Tesla's invention of Tesla coils introduced NFT technology that has been the foundation for many modern appliances and products today. Tesla coils achieve NFT by a method called inductive coupling; the wire coils induce a magnetic field allowing for electrical energy to be resonated to devices within the area of the field. There are, however, multiple methods to achieve NFT including resonant-inductive coupling which is the primary focus of this project as most consumer applications utilize this method.

The second focus area of this project is applications of far-field transmission (FFT). Contrary to NFT, FFT deals with transferring power over larger distances and uses a different technique called power-beaming and beamforming which projects electromagnetic radiation in a specific direction allowing it to travel greater distances. As promising as FFT sounds, it has presented researchers with several roadblocks. Issues such as interference, efficiency, and system cost all currently limit how viable this technology is. This project will showcase the construction of a typical resonant-inductive NFT system and then explain the advantages and current challenges associated with FFT systems.

I. Problem(s) Statement

Wireless Power Transmission (WPT) or *Wireless Energy Transmission* (WET) is the transmission of electrical energy without the use of a physical link [1]. WPT is a relatively new technology in the commercial marketplace where the first patent of its kind was created in 2009 [1]. The first WPT system, however, was created in 1891 by Nikola Tesla who developed *inductive coupling* Tesla coils

WPT serves as the generic name for a wide variety of technologies that transmit energy. These technologies can be sorted into two classes that encompass all WPT systems: *Near-field region* or *nonradiative region* (for *near-field transmission;* NFT) and *far-field region* or *radiative region* (for *far-field transmission;* FFT). While there are differences between the two, both have similar principles [2]:

1. A transmitter device is connected to a power source.
2. The transmitter converts power to a time-varying electromagnetic field.
3. A receiver device receives energy from the induced field.
    a. (Optional) the electromagnetic energy is converted to AC or DC via a rectifier.
4. A load is driven by this energy.

NFT and FFT are characterized by the distances at which a load can receive energy. The *energy transmission distance* (ETD) is completely dependent on the properties of the system being used. ETD is loosely characterized as there are neither set scientific nor industry standards for this parameter however it is typically broken up into the following [3][4]:

1. **Short Range:** ETD up to one diameter of the antenna (NFT systems)
2. **Mid-range:** ETD up to 10x diameter of the antenna (NFT and some FFT systems)

3. **Long Range:** ETD of greater than one wavelength of the antenna (FFT systems)

"Antenna" in the parameters above is used to indicate the type of transmitter being used. In the case of NFT systems, "antenna" refers to two coils coupled together to form an oscillating electromagnetic field. In the case of FFT systems, "antenna" is an actual antenna being used to beamform electromagnetic waves.

Looking at the parameters for short and mid-range ETD (thus NFT ETD), one can't help but wonder what this distance is. From a 2020 industry evaluation by the *Wireless Power Consortium*, companies producing wireless charging phone pads reported an average operating distance of 32-50 mm with an average coil diameter of 41.5 mm [5]. While the hassle of wires has been eliminated (not counting the plug-in of the actual charger), the phone or load for any NFT system in the short to medium range cannot cover more than half a centimeter in practical consumer applications [2].

This system limitation has driven researchers to look for more convenient ways to power your device by using a transmitter of greater energy field proximity. Without knowing anything more than its long-range capabilities, FFT systems should theoretically be able to solve this problem. Unfortunately, this is currently not the case. While there exists applications of FFT systems, most are for government or industrial use and not viable for consumer applications.

The scope of this project will involve discussion on the limitations of FFT systems in commercial settings and the construction of a modern-day NFT system used to power a simple load, a light emitting diode (LED). The design will be following Qi standards set by the Wireless Power Consortium to properly optimize, construct, and measure parameters for my NFT system.

II. Theory

As previously mentioned, WPT systems are sorted into two classes, NFT and FFT. NFT and FFT systems are further subdivided according to their ETD. NFT systems are classified as short to medium range and FFT systems are classified as long range. Figure 2 provides a breakdown of the current systems with corresponding ETD, frequency ranges, and antenna devices.

System	Range	Operating Frequency	Antenna Device	Class	Applications
Inductive Coupling	Short $\leq d_{antenna}$	Hz – MHz	Coils	NFT	Electric toothbrushes, metal detectors
Resonant Inductive Coupling	Medium $\leq 10 \cdot d_{antenna}$	kHz – GHz	Coils	NFT	Phone charging pads, implanted medical devices, electric vehicles
Capacitive Coupling	Short $\leq d_{antenna}$	kHz – MHz	Metal plates electrodes	NFT	RIFD smartcards, charging portable devices
Magnetodynamic Coupling	Short $\leq d_{antenna}$	Hz	Rotating magnets	NFT	Electric vehicles, biomedical implants
Microwaves	Long $\geq \lambda_{antenna}$	GHz	Parabolic dishes, phased arrays, rectennas	FFT	Drones, satellites, space applications
Light Waves	Long $\geq \lambda_{antenna}$	$\geq$ THz	Lasers, photocells, lenses	FFT	Drones, aircraft, space applications

Figure 2: *Classification of various WPT systems*

Near-Field Transmission Systems

As shown in figure 2, a commonality between all NFT systems is their shorter-range ETD relative to FFT systems. This occurs because the fields produced by NFT systems decrease with the cube of the distance $E, B = (\frac{d_{range}}{d_{antenna}})^{-3}$[6]. The transmitted power is proportional to the square of the field strength. Therefore, the power drop-off for increasing distances is $P = (\frac{d_{range}}{d_{antenna}})^{-6} \cong 60dB/decade$ [6]. From these relations, twice the distance between a transmitter and receiver results in a power decrease of $2^6 = 64$. NFT systems use non-radiative techniques (electric fields and magnetic fields) which is why the field strength decreases exponentially with increasing distance [6].

Maximizing the ETD and range is the goal of any WPT system, and the following are two common systems that do this. This project uses the following systems' properties for its design (specifically, the *Resonant Inductive Coupling* system). Other NFT systems mentioned in figure 2 will be omitted from this paper as they use different techniques than the two below.

Inductive Coupling [2], [7]

Inductive coupling systems transmit power by an induced magnetic field between two wire coils. Together, a transmitter and receiver coil form a *transformer*[1]. By *Ampere's Law*[2], an alternating current passes

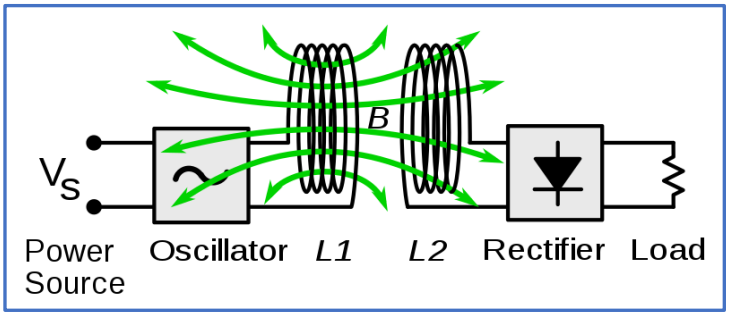


Figure 3: *Inductive Coupling System*

[1] A passive component that transfers electrical energy from one circuit to another
[2] Law describing the relation between a magnetic field around a loop and the current flowing through the loop

through the transmitter coil which induces a time-varying oscillating magnetic field. The induced magnetic field reaches the receiver coil where it creates an *electromotive force*[3] (EMF) by *Faraday's Law*[4]. The induced EMF will create an alternating current in the receiver coil which can drive a load or pass through a rectifier converting alternating current into direct current to drive a DC-dependent load (Figure 3). Most inductive coupling systems today use *electronic oscillators*[5] which produce higher frequency alternating currents. The higher frequency AC increases the total power transmission and overall efficiency [9].

The total power transmission also increases as the *mutual inductance* between the two coils increases. Mutual inductance describes how the voltage (EMF) induced in one coil affects the current change in a secondary coil. When the *magnetic flux*[6] is the same between both coils, the voltage induced in the first coil is proportional to the current change of the second coil [9]. The mutual inductance is then,

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{d} \qquad (\mathbf{1})$$

Where,

$\boldsymbol{\mu_0}$ = permeability of free space; $\boldsymbol{\mu_r}$ = permeability of material; $\boldsymbol{N_n}$ = turns of the "nth" coil

$\boldsymbol{A}$ = cross-sectional area; $\boldsymbol{d}$ = length between coils

Since *M* relies on the cross-sectional area, the orientation of the coils relative to each other affects the mutual inductance. Additionally, the length between the coils, *d*, will inversely

[3] Electrical work per unit charge on a charge carrier
[4] Law describing how a changing magnetic flux induces an electrical field and EMF
[5] Electronic circuit that produces an oscillating signal (sinusoidal, triangular, square waves, etc.)
[6] A measure of magnetic field lines passing through a given area

affect the mutual inductance [9]. Another way to describe the mutual inductance is by the inductances of the coils. If the inductances of the two coils are equal, the mutual inductance is described as,

$$M = \sqrt{L_1 L_2} = L \quad \textbf{(2)}$$

This would imply perfect coupling between the two coils. Of course, this is not possible and limitations like the coil separation and orientation cause an imperfect coupling. This error is represented by the coupling factor,

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad \textbf{(3)}$$

$$\rightarrow M = k\sqrt{L_1 L_2} \quad \textbf{(3.1)}$$

Where $0 \leq k \leq 1$; L_n = self-inductance of coil "n"

k is a dimensionless parameter equal to the ratio of magnetic flux through the transmitter coil that passes through the receiver coil when L_2 is open circuited. A value of $k = 1$ would indicate perfect coupling and all the total magnetic flux of the system would be unchanged (Figure 4). The coupling factor is related to the total system efficiency as well and will be discussed later in the *Procedure* section.

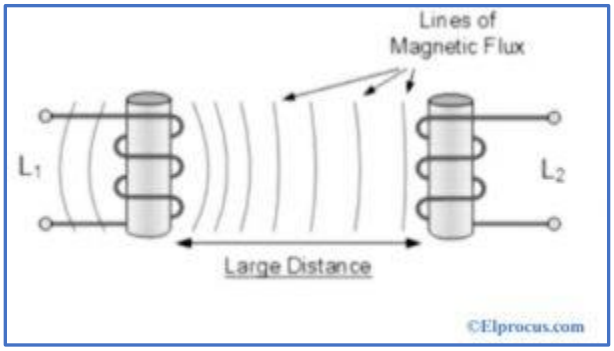


***Figure 4**: Inductive Coupling between two coils*

Even with the use of electronic oscillator circuits, coils in inductive coupling systems must be very close to each other in order to have optimal power transfer and system efficiency. This system can be enhanced by adding *resonant circuits* and is described in the next section.

Resonant Inductive Coupling [2], [8], [10]

The resonant inductive coupling system adds resonant (or tuned) circuits to the transmitter and receiver coils. Circuits that achieve resonance vary from capacitors connected to the coils, self-resonating coils, or other circuit configurations. The resonant circuits are tuned to

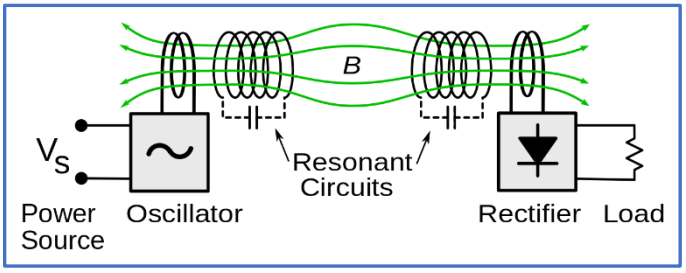


***Figure 5:** Resonant Inductive Coupling*

match the same *resonant frequency*. The resonance frequency is the naturally occurring frequency of the oscillatory circuit.

Resonance frequency occurs when the *inductive reactance*[7] is equal to the *capacitive reactance*[8]. That is,

$$X_L = X_C$$

Where,

$$X_L = 2\pi fL;\ \ X_C = \frac{1}{2\pi fC}$$

Thus,

$$2\pi fL = \frac{1}{2\pi fC}$$

And now, rearranging for resonance frequency

$$2\pi f^2 L = \frac{1}{2\pi C} \rightarrow f^2 = \frac{1}{4\pi^2 LC}$$

$$\rightarrow f = \frac{1}{2\pi\sqrt{LC}} \qquad \textbf{(4)}$$

When the total system inductance and capacitance is known, the resonance frequency can be determined. This is the operating frequency at which the highest power transmission is possible with the most efficiency.

[7] The opposition of the flow of current in an AC circuit by an inductor (measured in Ohms)
[8] The opposition of the flow of current in an AC circuit by a capacitor (measured in Ohms)

Resonant inductive coupling is an advancement of traditional inductive coupling, but the same principles of inductive coupling apply. The same parameters used to determine the overall power transfer and efficiency still apply as well. For resonant inductive coupling, we use one additional parameter to characterize the system efficiency which is the *quality factor* (denoted as *Q*). *Q* is a dimensionless parameter that describes the ratio of energy stored in a resonator to energy lost after one oscillation. *Q* is defined in terms of the resonance frequency,

$$Q = \frac{X_L}{R} = \frac{2\pi f L}{R}. \quad \textbf{(5)}$$

For high *Q*-factors, the inductance needs to be as large as possible and the total system resistance to be as low as possible. The total *Q* of a system is the geometric average of Q_1 and Q_2 and given by

$$\sqrt[n]{Q_1 \cdot Q_2 \cdot \ldots \cdot Q_n} \quad \textbf{(5.1)}$$

Where *n* defines the number of quality factors of the system (for the system design in this paper, it will just be two).

For resonant inductive coupling, the product of *Q* and *k* (Equ. 3) is the *figure of merit*[9] and is inversely proportional to the *loss factor* (λ) of the system, which is given by,

[9] A numerical characterization of the performance or efficiency of a system

$$\lambda = \frac{P_{loss}}{P_{out}}. \quad \textbf{(6)}$$

Figure 6 shows a figure of merit vs. loss factor which is applicable to all resonant inductive coupling systems. This is one way to determine the efficiency of a system and for the purposes of this project will serve as way to properly optimize the system before making further enhancements.

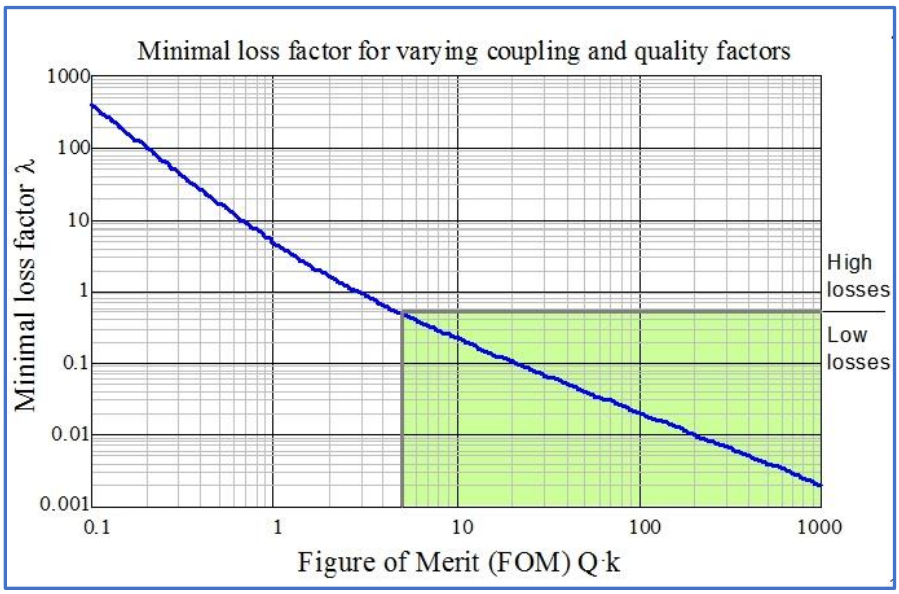


***Figure 6**: System Loss Factor for Inductive and Resonant Inductive Coupling Systems as defined by the Wireless Power Consortium. The "green zone" indicates acceptable values of low loss for a system.*

Scattering Parameters (S-Parameters) [11], [12]

Another way to measure the system efficiency is by measuring its scattering parameters. S-parameters describe the behavior of linear electric networks when undergoing change from electrical signals. They are often used in *two-port networks*[10] to analyze values such as return

[10] An electrical circuit with two terminals that connect to external circuits

loss, insertion loss, and system efficiency. Because a resonant inductive coupler behaves as a two-port linear network, measuring the scattering parameters are very useful in determining the power transfer and system efficiency.

An *n*-port system is characterized by a square matrix of complex numbers referred to as the S-parameter matrix (Figure 7.1). The matrix is used to calculate its response to input signals at its ports. For a generic *n*-port system, the S-parameter matrix is defined in terms of its incident and reflected *power waves*[11]. The incident and reflected power waves respectively are given as,

$$a_n = \frac{1}{2} k_n (V_n + Z_n I_n) \qquad \textbf{(Equ. 7)}$$

$$b_n = \frac{1}{2} k_n (V_n - Z_n^* I_n) \qquad \textbf{(Equ. 8)}$$

Where,

$\boldsymbol{k_i} = (\sqrt{|R\{Z_i\}|})^{-1}$; $\boldsymbol{Z_i}$ = impedance at port "*i*"; $\boldsymbol{Z_i^*}$ = complex conjugate of Z_i; $\boldsymbol{V_i}$ = complex amplitude of voltage; $\boldsymbol{I_i}$ = complex amplitude of current

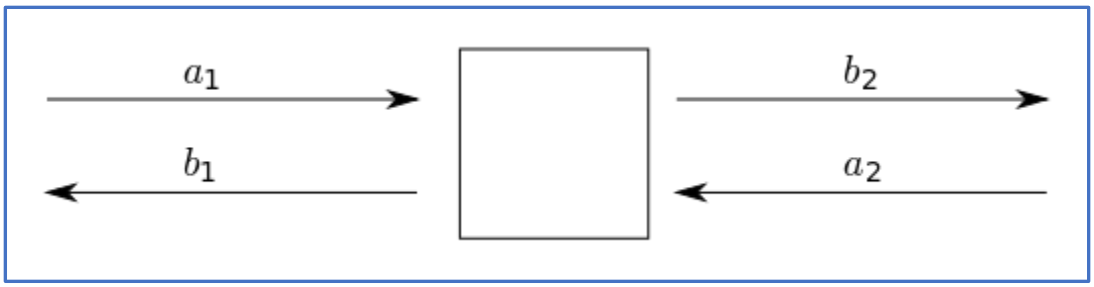


***Figure 7**: Two-Port Network of incident a_1 and a_2 power waves and reflected b_1 and b_2 power waves*

[11] Waves of AC signal generated at resonance frequency

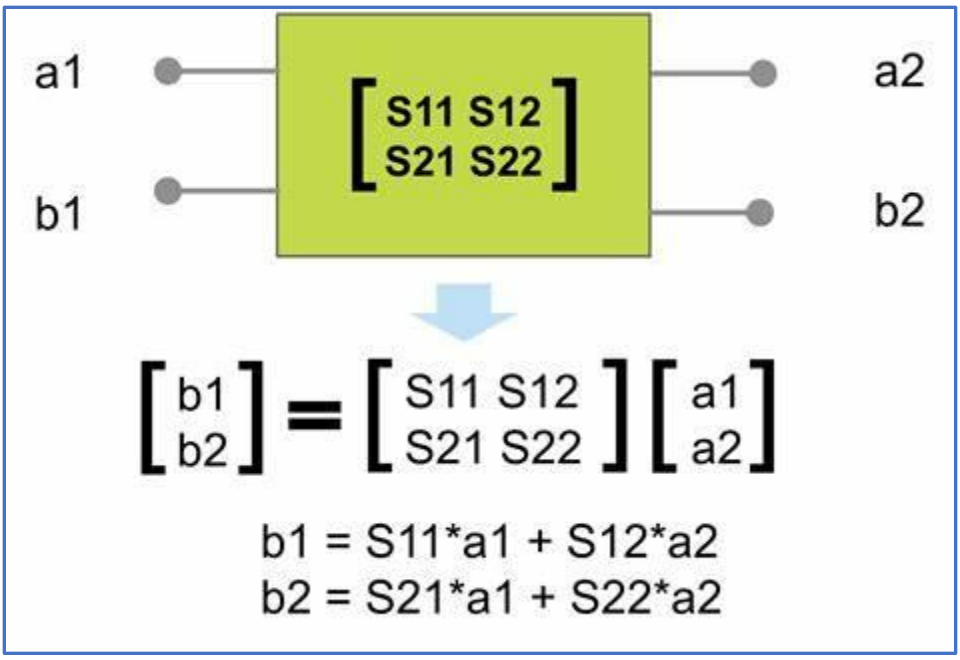


Figure 7.1*: Example of a two port S-parameter matrix*

We can then rewrite equation 7 and equation 8 as,

$$a_n = \frac{1}{2}\frac{(V_n + Z_n I_n)}{\sqrt{|R\{Z_i\}|}} \quad \textbf{(9)}$$

$$b_n = \frac{1}{2}\frac{(V_n - Z_n^* I_n)}{\sqrt{|R\{Z_i\}|}} \quad \textbf{(9.1)}$$

For a two-port system, as in our case, the S-parameter matrix looks like,

$$\begin{matrix} b_1 \\ b_2 \end{matrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad \textbf{(10)}$$

Where,

a_1 = primary incident power wave; a_2 = secondary incident power wave; b_1 = secondary reflected power wave; b_2 = primary reflected power wave

Expanding the matrix gives the equations

$$b_1 = S_{11}a_1 + S_{12}a_2 \quad \textbf{(11)}$$

$$b_2 = S_{21}a_1 + S_{22}a_2 \quad \textbf{(11.1)}$$

These equations show the relation between the incident and reflected power waves at each port of the system which in this case is the input of the transmitter coil and the output of the receiver coil. The definitions for each scattering parameter are,

- $\boldsymbol{S_{11}}$ = input port voltage reflection coefficient
- $\boldsymbol{S_{12}}$ = reverse voltage gain
- $\boldsymbol{S_{21}}$ = forward voltage gain
- $\boldsymbol{S_{22}}$ = output port voltage reflection coefficient

Once the S-parameters are measured, there are several values that can be determined to understand the efficiency or effectiveness of the system. Additionally, some of these parameters do not measure efficiency in the standard sense and that is why the terminology "effectiveness" is used also.

1. **Complex Linear Gain**

   $$G = S_{21} = \frac{b_2}{a_1}$$

2. **Scalar Linear Gain**

$$|G| = |S_{21}|$$

3. **Scalar Logarithmic Gain**

   $$G = 20\log_{10}|S_{21}|\ dB$$

4. **Insertion Loss**

   $$IL = -20\log_{10}|S_{21}|\ dB$$

5. **Input Return Loss**

   $$RL_{in} = -20\log_{10}|S_{11}|\ dB$$

6. **Output Return Loss**

   $$RL_{out} = -20\log_{10}|S_{22}|\ dB$$

Electromagnetic Field Theory [10]

Scattering parameters are the most optimal way to determine the efficiency of a WPT system and almost always used for this purpose. With the use of a vector network analyzer (VNA), the S-parameters can be easily measured at the input of the transmitter and output of the receiver. Beyond this purpose, a VNA does not have other useful applications making it an uncommon and highly expensive piece of equipment. Lower grade VNAs range anywhere from $1,000 - $3,000. This is a limitation on the project and requires the use of electromagnetic field theory to determine the efficiency of the system.

Having a maximum system efficiency or *power transfer efficiency* (PTE) is a primary objective of this project. From a lengthy derivation involving EM-field theory, we derive an equation for PTE,

$$PTE = \frac{P_{out}}{P_{in}} = \frac{P_2}{P_1} = \frac{|I_2|^2 R_L}{|I_1|^2 R_1 + |I_2|^2 R_2 + |I_2|^2 R_L}$$

Which can be simplified as

$$PTE = \frac{\omega^2 M^2 R_L}{R_1[(R_2+R)^2+(\omega L_2+X)^2]+\omega^2 M^2 (R_2+R)} \quad \textbf{(12)}$$

The PTE is a ratio of the output power to input power and is the general Qi standard for measuring system efficiency by the Wireless Power Consortium. For clarity, it is best to work backwards from this equation to better understand how and what these variables are.

The output power $P_{out} = P_2$ is the power provided to the load from the receiver. The input power $P_{in} = P_1$ is the total power generated by the system. These are both given as,

$$P_2 = |I_2|^2 R = |I_1|^2 \frac{\omega^2 M^2 R}{(R_2+R)^2+(\omega L_2+X)^2} \quad \textbf{(13)}$$

$$P_1 = |I_1|^2 R_1 + |I_2|^2 (R_2 + R) = |I_1|^2 (R_1 + \frac{\omega^2 M^2 (R_2+R)}{(R_2+R)^2+(\omega L_2+X)^2}) \quad \textbf{(14)}$$

It is important to note that these derivations hold true for a coupling system that uses series matching on the primary side (that is a capacitor in series with the transmitter) and parallel matching on the secondary side (capacitor in parallel with the receiver). This is an important distinction and will not hold true if the system has a different configuration [10]. For parallel matching on the secondary side, we have the following values,

$$R = \frac{R_L}{\left(\omega C_{2_p} R_L\right)^2 + 1} \quad \textbf{(15)}; \quad X = \frac{-\omega C_{2_p} R_L^2}{(\omega C_{2_p} R_L)^2 + 1} \quad \textbf{(16)}$$

$$R_L = R + \frac{1}{R\left(\omega C_{2_s}\right)^2} \quad \textbf{(17)}; \quad \omega C_{2_p} = \frac{-X}{R^2 + X^2} \quad \textbf{(18)}$$

Additionally, the relationship between the two currents is

$$I_2 = \frac{j\omega M}{R_2 + j\omega L_2 + R + jX} I_1 \quad \textbf{(19)}$$

Where *j* is the imaginary unit $\sqrt{-1}$

Using the following derivations (equations 12-19), one can find the theoretical *PTE* by measuring the input current, frequency, load resistance, and selecting a proper capacitor for the receiver and transmitter coils. The project is now ready for construction.

III. Objective(s)

1. Maximize Total *PTE*

My resonant inductive coupling system will be designed according to standards set by the Wireless Power Consortium. The Wireless Power Consortium is a multinational consortium comprised of WPT manufacturers that sets *Qi* interface standards. *Qi* standards are set for materials within a system, acceptable power transfer ratings (including *PTE*), and methods/criteria for measuring power transfer ratings. By following guidelines in place by the Wireless Power Consortium and the useful derivations from EM field theory, this project's goal is to have a *PTE* comparable to commercial WPT systems and according to the minimum standard set by Wireless Power Consortium.

2. Demonstrate Energy Drop-off

Even with the proper optimizations and guidelines, inductive coupling systems (even with resonant circuits) cannot transmit sufficient power for a practical application. The open interface *Qi* standards defined WPT for inductive systems of distances up to 4 cm. [15] Through observation and experimentation by industry manufacturers, the most sophisticated systems can transmit reasonable amounts of power up to this distance. There are inductive systems that can transmit power beyond this distance, however, the size of the system must become much larger (coils of >0.25m diameter) which doesn't make it practical for consumers. After construction, the design will demonstrate how and why this energy drop-off occurs.

3. Explaining Far-Field Transmission

The final objective will be explaining the active research within FFT systems, the current limitations holding back its anticipated applications, and the anticipated applications themselves.

IV. Procedure

The first part of the design process involved purchasing two identical *Onn.*^{TM} *10W* commercial charging pads. These charging pads use resonant inductive coupling technology that are manufactured according to *Qi* standards (Figure 7). The charging pads use coils outsourced by Würth Elektronik Corporation. After salvaging the coils, the data sheet was referred to that lists useful values including *Q-factor vs. Frequency* (Figure 8).



Figure 7: *Onn. Wireless Charging Pad*

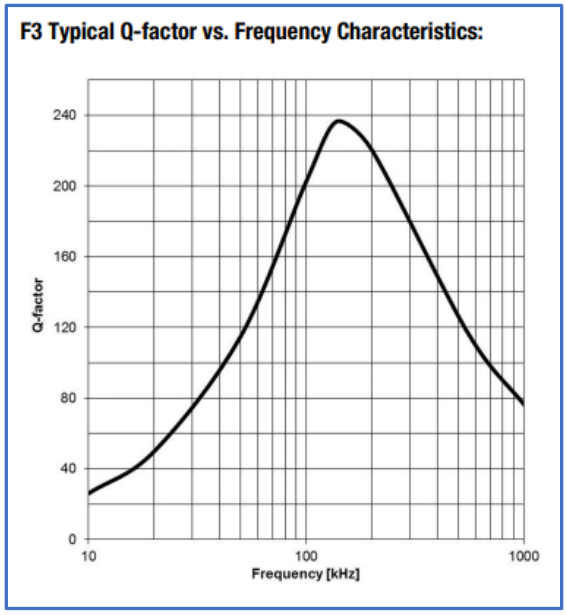


Figure 8: *Q-Factor vs. Frequency*

The system was then optimized to operate at a maximum efficiency before connecting the PWM circuit. The methods for this optimization follow in the next several paragraphs

Referencing back to figure 6, the power loss is inversely related to the figure of merit $Q \cdot k$. The coupling factor cannot be affected (except by increasing the distance/coil orientation) however the coils' quality factor Q can be influenced. Using equation 5, the inductance can be maximized, and the resistance minimized in order to have a high Q. Commercial coils were salvaged because they do both very well. Commercial coils use hundreds of small thin copper

wires wound together to minimize resistance by avoiding the *skin effect*. The skin effect is the tendency of an AC to become distributed within a conductor such that the current density is largest near the surface. The idea is that thicker wire will have too much resistance for the current to penetrate its inner radius and have a non-conductive zone (Figure 9). Ideally, the *skin depth* (δ) should be greater than the diameter of the coil (Equation 20).

(20) $$\delta = \sqrt{\frac{2\rho}{\omega\mu}}\ ;$$

ρ = resistivity of copper
ω = angular frequency of current
μ = permeability of copper

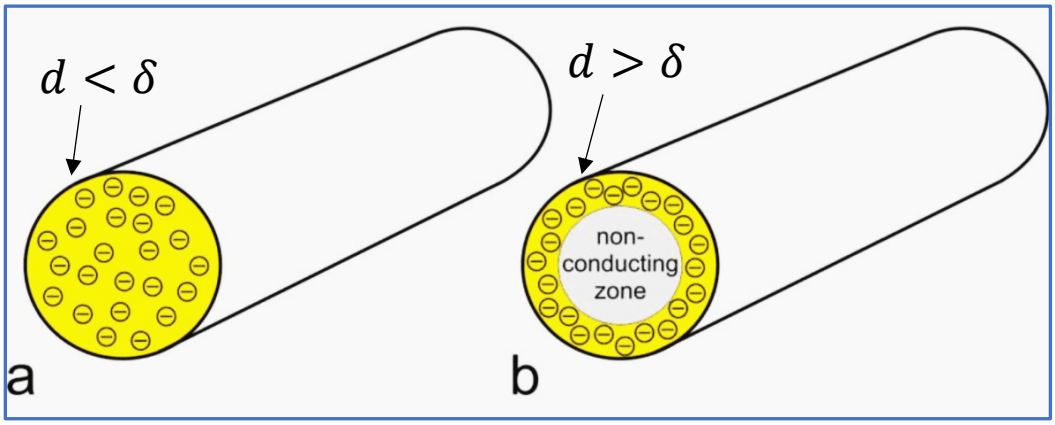


Figure 9: *Object "a" with a skin depth greater than the coil diameter (total conductance) and object "b" with a skin depth less than the coil diameter (non-conducting area)*

The inductance of the coil can be increased by creating more windings. More windings, however, would cause increased resistance because the length is increasing. Industry leaders have determined a "sweet spot" for the number of windings to have a high inductance while as not to increase the resistance too much [13]. Additionally, most commercial coils use *ferrite plates* which is a type of non-conductive metamaterial that increases inductance in surrounding components. Ferrite plates do this by suppressing *eddy currents*[12] which better focuses magnetic

[12] Currents that flow perpendicular of the induced magnetic field indicating leakage or thermal energy loss

fields. After attempting to test the inductance values with and without ferrite plates on my coils, there were some issues. The ferrite plates are superglued very tightly to the coils and extremely difficult to take off without breaking. Ferrite plates lose effectiveness the more the material is fractured. Therefore, the experiment was performed using separate coils and identical ferrite plates to see if they increase coil inductance. The results for that test are as follows:

Inductance without Ferrite Plate

$$L = \frac{V}{2\pi If} = \frac{0.53V}{2\pi(0.0189A)(125kHz)} = 55.2\mu\text{H}$$

Inductance with Ferrite Plate

$$L_f = \frac{V}{2\pi If} = \frac{0.82V}{2\pi(0.0173A)(125kHz)} = 60.4\mu\text{H}$$

So, the ferrite plates do help to increase the inductance as expected. After performing this test, the self-inductances were measured for both coils at resonance frequency (which will be the same as they are identical) which came out to be 22.1 μH. The resonance frequency comes from the coil's data sheet which lists a value of 125kHz (Figure 10). This appears accurate as the specified value is 24 μH at resonance.

D Electrical Properties:

Properties	Test conditions		Value	Unit	Tol.
Inductance	125 kHz/ 10 mA	L	24	µH	±10%
Q-factor	125 kHz/ 10 mA	Q	180		typ.
Rated current	ΔT = 40 K	I_R	6.0	A	max.
Saturation current		I_{sat}	10.0	A	typ.
DC Resistance	@ 20°C	R_{DC}	0.07	Ω	typ.
DC Resistance	@ 20°C	R_{DC}	0.1	Ω	max.
Self resonant frequency		f_{res}	5	MHz	

Figure 10: *Coil Electrical Properties*

After measuring the inductance, the figure of merit can be calculated to determine the power loss of the system. If the power loss is within the green area from figure 6, the system can move

forward in the construction process. The results of that calculation will be shared in the *Data* section however the system did meet this criterion.

The next phase of the design was to for the resonant circuit was to solder two capacitors (one in series with the transmitter coil and one in parallel with the receiver coil). As a result, the system acts as a low-pass LC-filter allowing for a resonance frequency. The capacitor values were determined by equation 4 where the resonance frequency and inductance are both known. The capacitor values came out to be 57.3 nF.

After soldering, 0.080” x 18” x 24” acrylic glass was used to mount the coils (Figure 11). The transmitter coil was mounted in the center of two 5”x5” acrylic sheets screwed together. The receiver coil was mounted in the center of two 4”x4” acrylic sheets screwed together. The thinnest acrylic sheets available were chosen to keep the separation between the coils at a minimum. An LED was then attached to the receiver to act as the load.

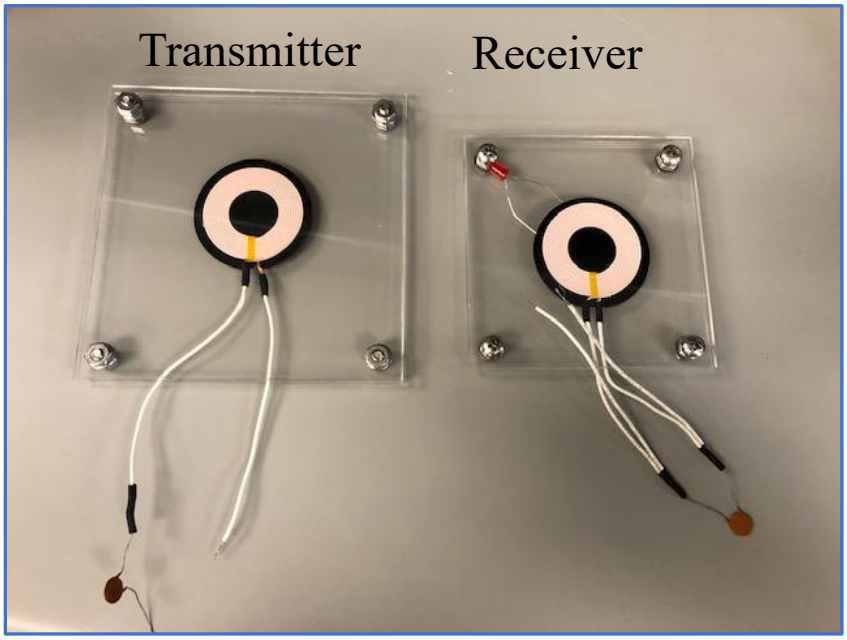


Figure 10: *Transmitter coil with capacitor in series and receiver coil with capacitor in parallel (Two Port Resonant Inductive Coupling System)*

A brief test was then conducted to ensure the system operated (Figure 11). The Hastings College logo was also placed on the receiver for free advertisement.

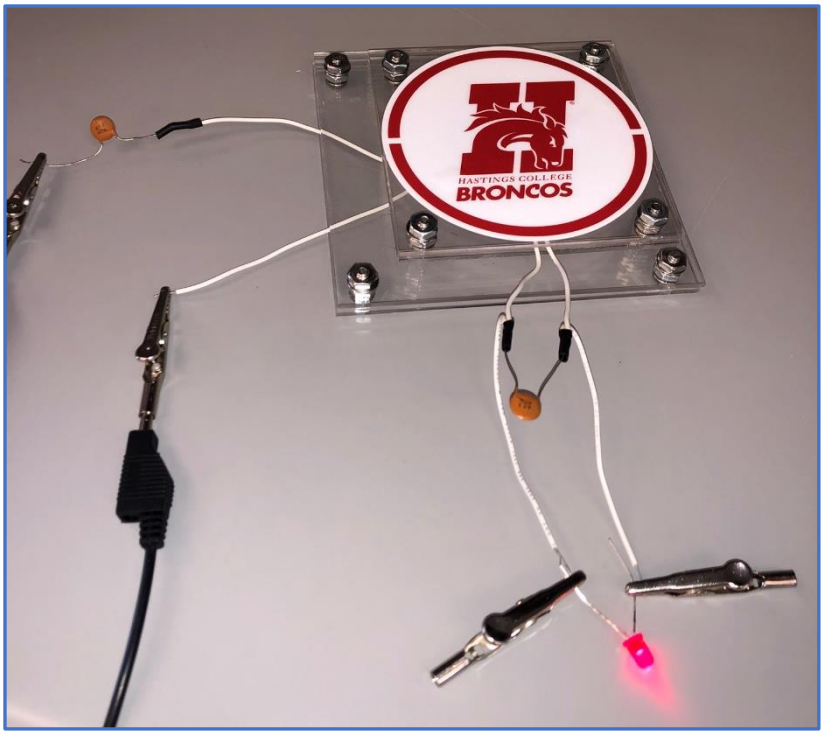


Figure 11: *Completed Resonant Coupling System*

The result is a completed resonant inductive coupling system. At this point, data can be acquired to determine the system's PTE and total power transfer by use of the derivations from EM field theory in the *Theory* section. The design was enhanced by adding a pulse-width modulation H-bridge circuit. After connecting the circuit, parameters necessary to calculate the PTE were measured.

Pulse-Width Modulation H-Bridge

An H-bridge circuit is a circuit that contains four switching elements. Typically, the switching elements are *transistors*[13]. The circuit also includes diodes called *catch diodes*. Their purpose is to mitigate *flyback*[14] common across inductive loads. In general, each transistor can be turned on and off independently. The transistor switches can be exchanged for relays instead so that only one control (a PWM control) is needed which is what this circuit does. The job of the H-bridge in this circuit is to translate the input voltages into amounts that the gates can drive.

The H-bridge is working with a pulse-width modulation (PWM) control in this circuit in order to digitally control the output current by changing the input voltage. The goal is to increase the current by means of the transistors rectifying DC to AC and pulse-width controlling its frequency. For a better understanding, figure 12 and 13 shows the components and their functions are explained on the following pages.

[13] A semiconductor device with three connections, capable of amplification in addition to rectification
[14] The sudden voltage spike seen across inductive loads when the supply voltage is reduced or removed

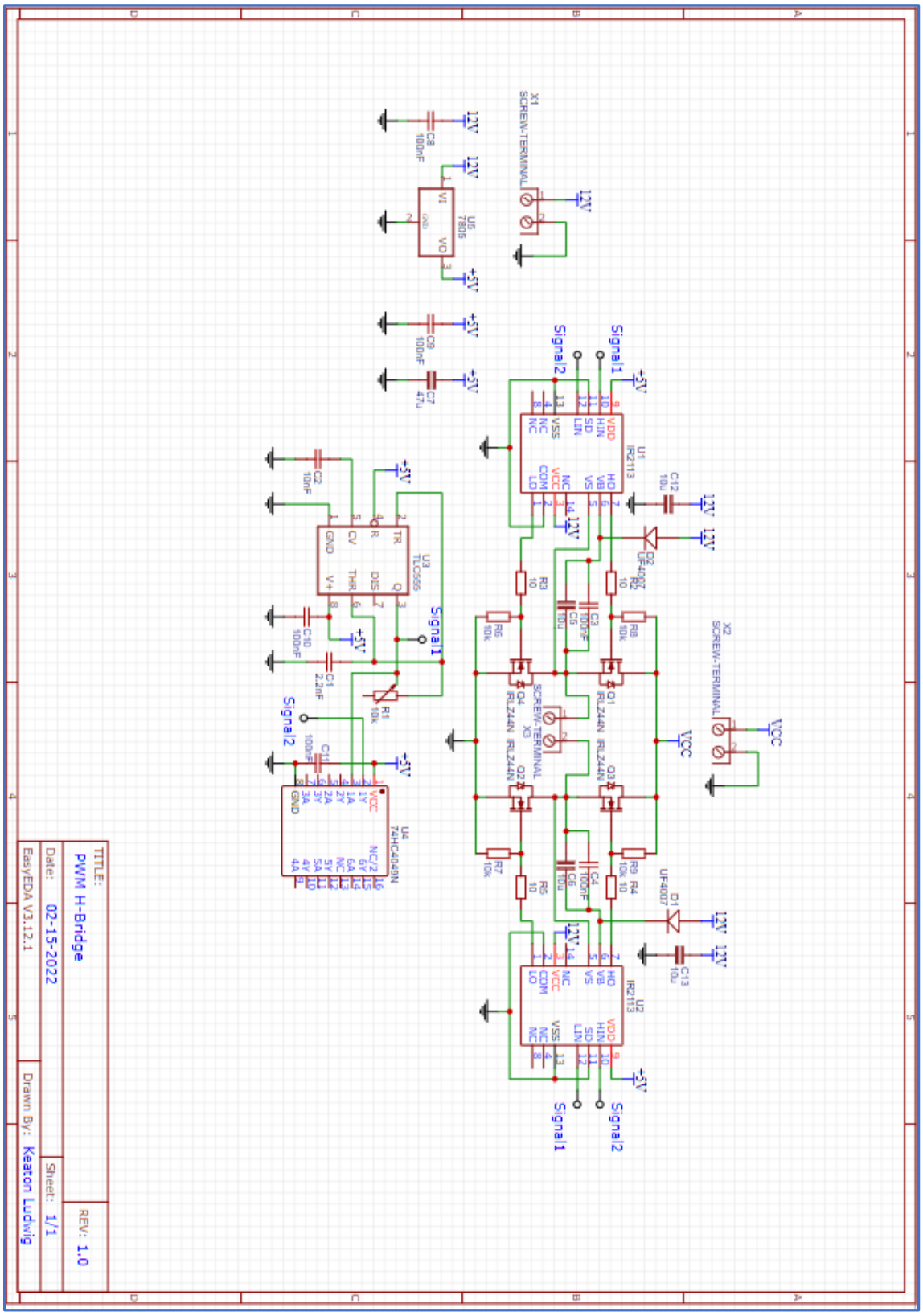


Figure 12: *PWM H-Bridge Schematic*

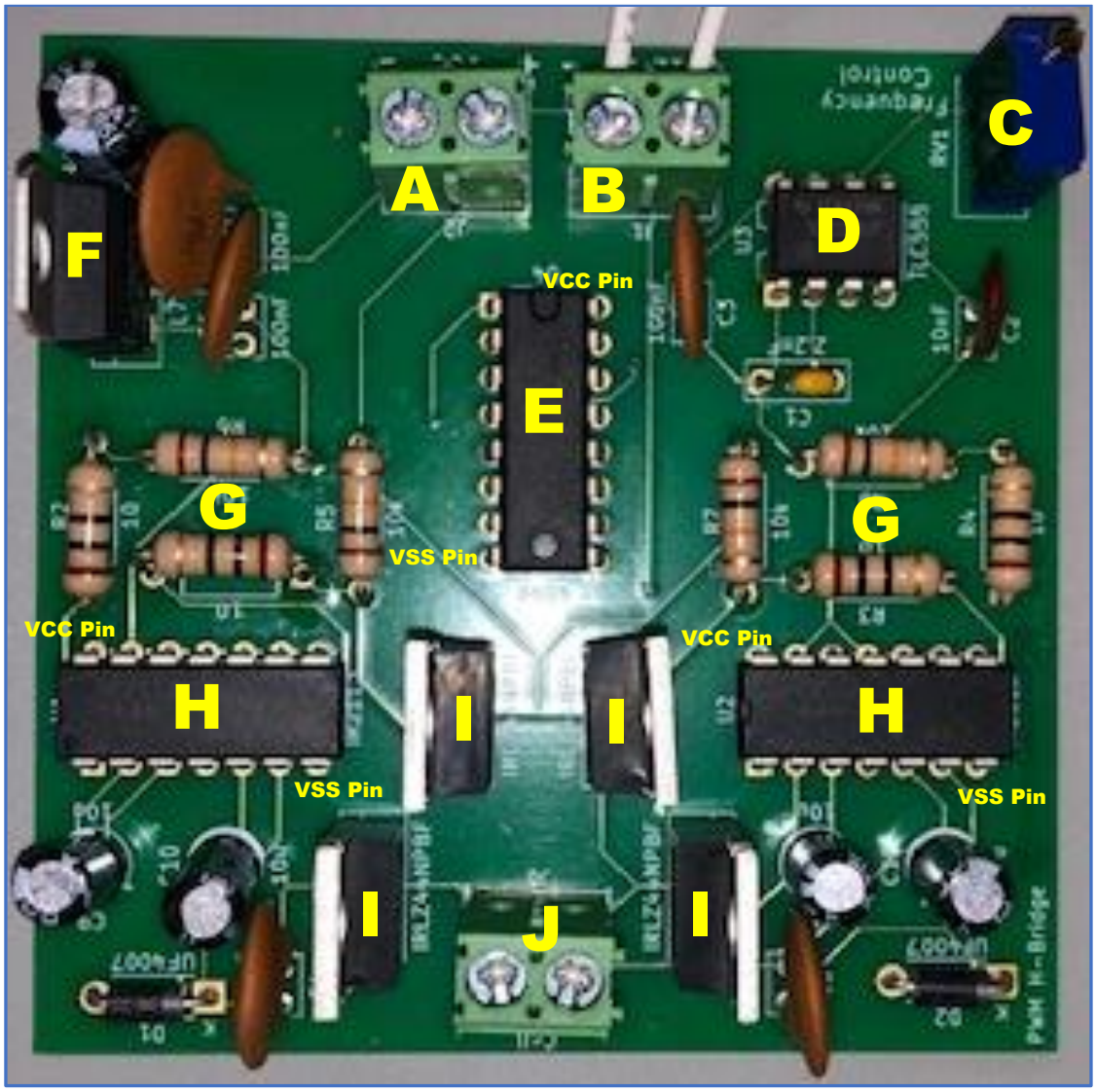


Figure 13: *PWM H-Bridge Circuit*

A. VCC Power Supply

Voltage Common Collector (VCC) is an input power supply different from the more commonly used *voltage source supply* (VSS). Unlike a VSS, a VCC can be either positive or negative with respect to ground and is used in circuits where the polarity of current flow must change. The VCC connects to the top right pins on **E** and **H** that are

designed to change polarities (the automatic switches mentioned above) so that the signal can flow through the entire circuit.

B. DC Power Supply

The processors **E** and **H** require a positive power supply input with respect to ground, so a 12V DC VSS supplies power to them on their bottom left pin locations. Most processors and microprocessors use the top right pin as a VCC input and bottom left pin as a VSS input.

C. 10 kΩ Potentiometer

This works in tandem with the TLC555 Timer. The potentiometer is the frequency control of the circuit by acting as a variable resistor. When installed, a flathead screwdriver is used to adjust the "trimmer" on the top to an appropriate frequency. This frequency is dependent on the resonant frequency of the system.

D. TLC555 Timer

The 555 timer (Figure 14) receives the signal set at the appropriate frequency by the potentiometer and then uses three internal transistors to divide the VCC voltage of the signal. The timer does this a certain number of times per second generating a pulse for the new signal. The pulse is also controlled by the

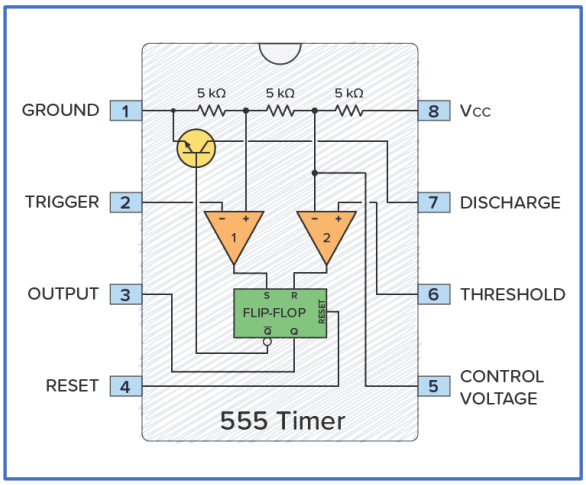


Figure 14: *555 Timer Diagram*

calibration of the trimmer on the potentiometer. The signal is then sent out to appropriate pins of **E** and **H** for further processing.

E. 74HC4049 Buffer Inverter IC

This component is what connects most everything in the circuit together (Figure 15). The pins on the buffer inverter have different functions but the two most important are the VCC input pin on the top right and VSS input pin on the bottom left. The VCC input pin receives the signal created by the potentiometer and 555 timer which then sends the signal to **F** on the left side of the bridge and **G** on the right side of the bridge for further processing. The VSS input pin simply receives power so the processor can function (usually at least 5V).

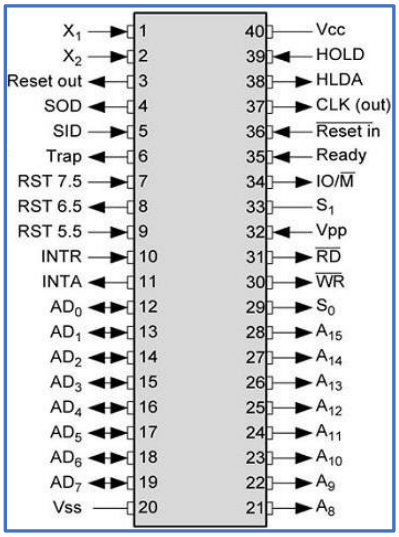


Figure 15: *Example of a Similar Processor*

F. L7805 Voltage Regulator

The voltage regulator is a component that maintains a constant unchanging voltage. This is useful in circuits that have VCC supplies where the polarity is changing. The voltage

regulator in this circuit receives the VCC signal from **E** and directs it towards **G** (on the left side).

G. Cap/Resistor Configuration

Two halves of the signal have now reached both sets of **G**, the top half of the PWM signal and the bottom half of the PWM signal. **G** is a configuration of resistors and capacitors that further tune the signal according to the needs of the system. The configuration acts as a filter and affects both halves of the signal identically.

H. IR2113 MOSFET Driver IC

MOSFET drivers are *metal-oxide-semiconductor field-effect transistors*. MOSFET drivers switch or amplify voltages in circuits. Their purpose in this circuit is to amplify the voltage allowing a low current input signal (around 0.2A for this system) to drive the MOSFET gates. They work together with the gates and create a compatible signal for the gates to rectify the signal into AC.

I. IRLZ44N MOSFET Gate

The two MOSFET gates on either side of the bridge receive an upper or lower half of the PWM signal from the drivers and then rectify the PWM DC signal to a sinusoidal AC signal. The rectification generates our desired AC signal and as a result more current can flow because of this signal conversion.

J. Terminal Output

The wire connections of the transmitter coil are fastened into the green screw terminal.

V. Project Management

As mentioned, not all materials listed were used for construction. The initial design was scrapped and redone to be compatible with the PWM H-Bridge circuit. The materials not used will be listed with an asterisk*.

Item	Supplier	Quantity	Unit Cost	Total
12 AWG Copper Wire (80ft)*	Wire & Cable Your Way	80	0.26	20.80
TS-400-124A Power Supply*	Digi-Key Electronics	1	138.44	138.44
Voltmeter	Hastings College	1	0.00	0.00
MOSFET N-Channel (>100V, >30A)*	Digi-Key Electronics	2	3.74	7.48
50µH Internal Resistor*	Amazon	2	17.95	35.90
120V Fast Diode*	Digi-Key Electronics	3	0.63	1.89
0.5µF MKP capacitor*	TTI	2	11.42	22.84
TAP105K035SCS 1µF capacitor*	TTI	1	0.38	0.38
228TTA035M 2200µF capacitor*	TTI	1	3.55	3.55
CD74HC4049E IC BUFFER INVERT 6V 16DIP	Digi-Key Electronics	1	0.65	0.65
IRLZ44NPBF MOSFET N-CH 55V 47A TO220AB	Digi-Key Electronics	4	1.52	6.08

IR2113PBF IC GATE DRVR HALF-BRIDGE 14DIP	Digi-Key Electronics	2	4.86	9.72
Electrolytic Capacitor Kit	Amazon	1	11.98	11.98
C315C222K1R5TA7303	Digi-Key	1	0.38	0.38
TLC555CP	Digi-Key	1	0.97	0.97
Custom PCB Board	Digi-Key	1	589.98	589.98
Skytopp VCC Power Supply	Amazon	1	69.00	69.00

Grand Total (excluding shipping): $920.04

The following Gantt chart was used as a timeline for the project.

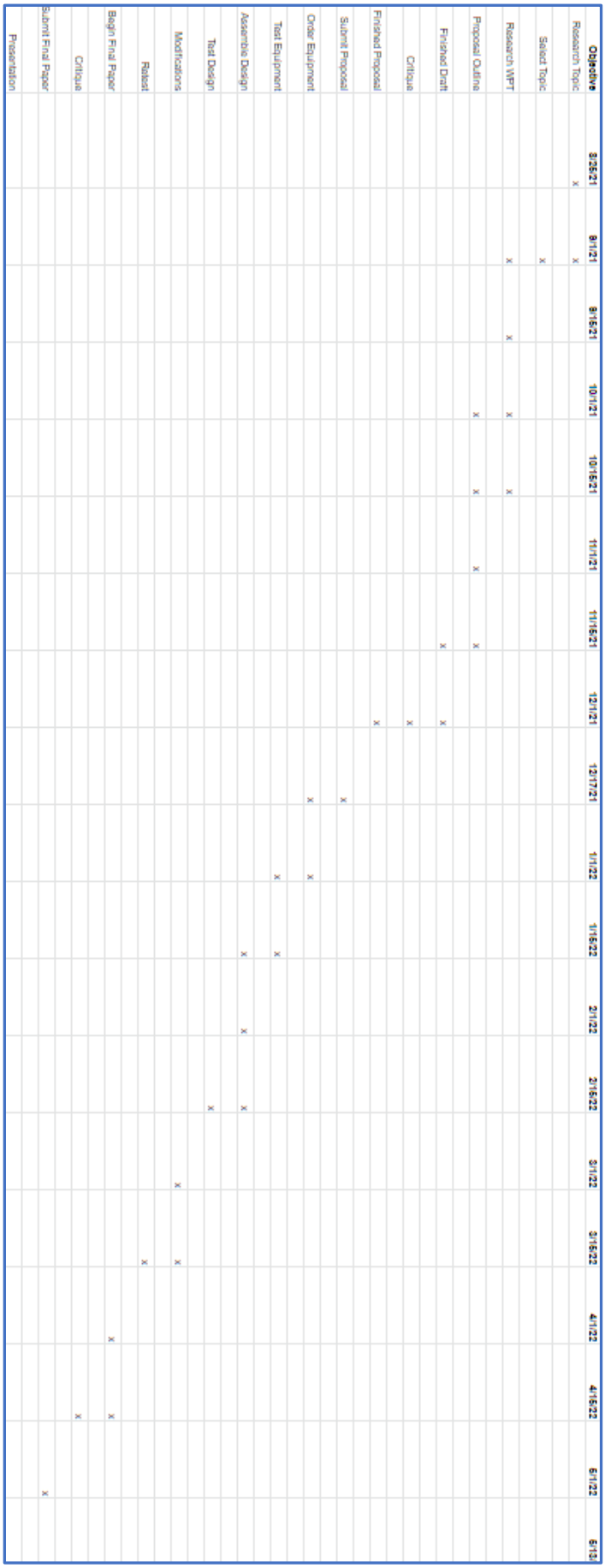


Figure 16: *Gantt Chart used to formulate a schedule for project deadlines*

VI. Data & Analysis

Optimization Results

The first phase of the design process was to optimize the system to have the highest possible PTE. This was done by increasing the figure of merit ($Q \cdot k$) where we can raise the value of *Q* by increasing *L* and decreasing *R* (equation 5). The coupling factor *k* was determined according to equation 3 after measuring the inductances. If the figure of merit is in the "green zone" then the loss factor is minimal for the system and further construction can commence. The following are the measured values for *Q* and *k*.

$$Q = \frac{X_L}{R} = \frac{2\pi f L}{R} = \frac{2\pi(125\ kHz)(22.1\mu H)}{0.107\Omega} = 162.2$$

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{4.47\mu H}{\sqrt{(22.1\mu H)^2}} = 0.202$$

$$Q \cdot k = 32.76$$

From figure 6, we get a minimal loss factor of approximately $\lambda = 0.08$ which is in the green zone indicating low levels of power loss. Additionally, the calculated $Q = 162.2$ seems to be accurate when compared to the manufacturer's specified value of $Q = 180$ at resonance. The system is now optimized according to *Qi* standards and further construction can begin.

Oscilloscope Measurements

After soldering the capacitors and connecting the PWM circuit, an output signal was taken before and after the drivers were installed to ensure the circuit operated correctly. Without the drivers installed, the output signal of the circuit to the transmitter should be an un-rectified square-wave DC signal (Figure 17). After verifying this, the drivers were installed, and the new signal was a sinusoidal AC as desired (Figure 18). The voltage used for both power supplies (VCC and VSS) were 12 V and the VCC's adjustable current supply was 0.202 A.

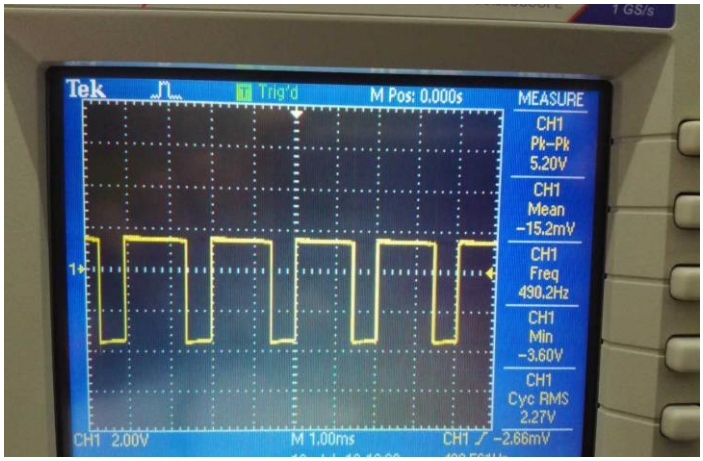


Figure 17: *Unrectified signal from the 12V DC power supply without VCC supply or MOSFET drivers installed*



Figure 18: *12V DC power supply signal (blue) and VCC rectified AC signal (pink) after MOSFET driver installation*

Power Transfer Efficiency (PTE)

The PTE is the most important aspect of the system. There is no one required measurement in place by the Wireless Power Consortium for a system's efficiency as there are many different parameters for this (as described on page 15) and completely dependent on the objective of the system. Typically, however, for inductive coupling systems, it is generally advised according to *Qi* standards set by the Wireless Power Consortium, that efficiency measurements for inductive coupling systems is $\frac{P_{out}}{P_{in}}$ as described by equation 12. Using this equation, the theoretical PTE is

$$PTE = \frac{P_{out}}{P_{in}} = \frac{\omega^2 M^2 R_L}{R_1[(R_2 + R_L)^2 + (\omega L_2 + X)^2] + \omega^2 M^2 (R_2 + R_L)}$$

$$= \mathbf{82.03}\%$$

This efficiency seems to be up to par with commercial charger manufacturers. According to the Wireless Power Consortium, the *Qi* efficiency standard is set at $\geq 85\%$. While the PTE of my system just misses this mark, the design can be still be viewed as a success. Large corporations use advanced integrated circuits with microprocessors that much better optimize inductive coupling systems than my larger design.

Efficiency Dropoff

The results of this project are meant to show how PTE decreases as the coupling factor decreases. The coupling factor, *k*, is dependent on the distance between the coils and the orientation of the coils with each other. A perfect orientation with no separation would indicate perfect coupling and a value of *k* that approaches 1. To show how *k* affects the PTE, the orientation was kept the same and the distance slowly increased by 0.1cm increments where *k* was then determined by measuring the mutual inductance *M* and then determining the PTE by equation 12. The results of that experiment are shown in figure 19.

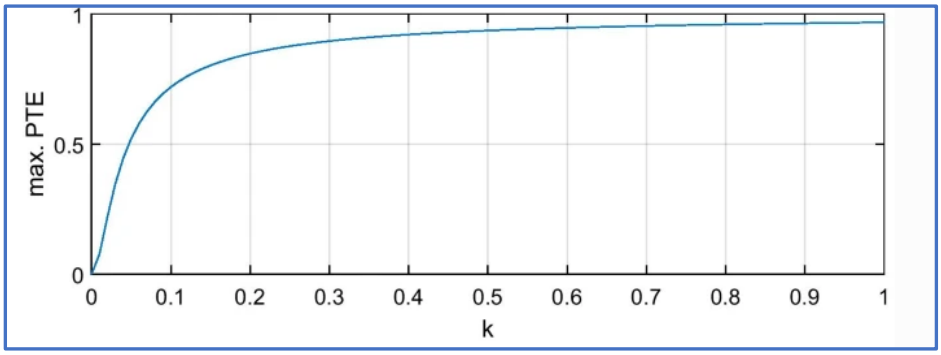


Figure 19: *PTE vs. Coupling Factor*

As expected, the theoretical PTE decreases quickly at coupling factors of approximately 0.2. The distance at which no light was observed by the LED was approximately 3.20 cm. This seems reasonable in comparison to the maximum 4 cm distance that resonant inductive coupling systems are supposed to be able to transfer sufficient power at (for the purposes of charging a device).

VII. Conclusions

The WPT project was successful in transmitting energy to power a small LED at an efficiency that is acceptable. The PTE of the system does not conform to the *Qi* standard however this standard is in place for large corporations that have large fundings and access to advanced integrated circuits and very small microprocessors. The fact that this system design was a small percentage outside the accepted range is acceptable considering the very different nature of the design when compared to large manufacturers.

The addition of the PWM circuit was extremely beneficial to the project. This enhancement allowed for a greater amount of current to flow into the transmitter thus increasing the total PTE. Without this circuit, the PTE would have been far less. WPT manufacturers include similar circuits in their systems to increase the total power output and PTE. These circuits are slightly different than the one used for this system because they use USB-C (or similar) inputs and micro-components.

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